Manufacturing Process Executive Summary

Key Performance Indicators

Simulation LOTs: 5,000

Normal LOTs: 4,321 (86.42%)

Defect LOTs: 679 (13.58%)

Overall Average Yield: 85.11%

Normal LOT Average Yield: 98.48%

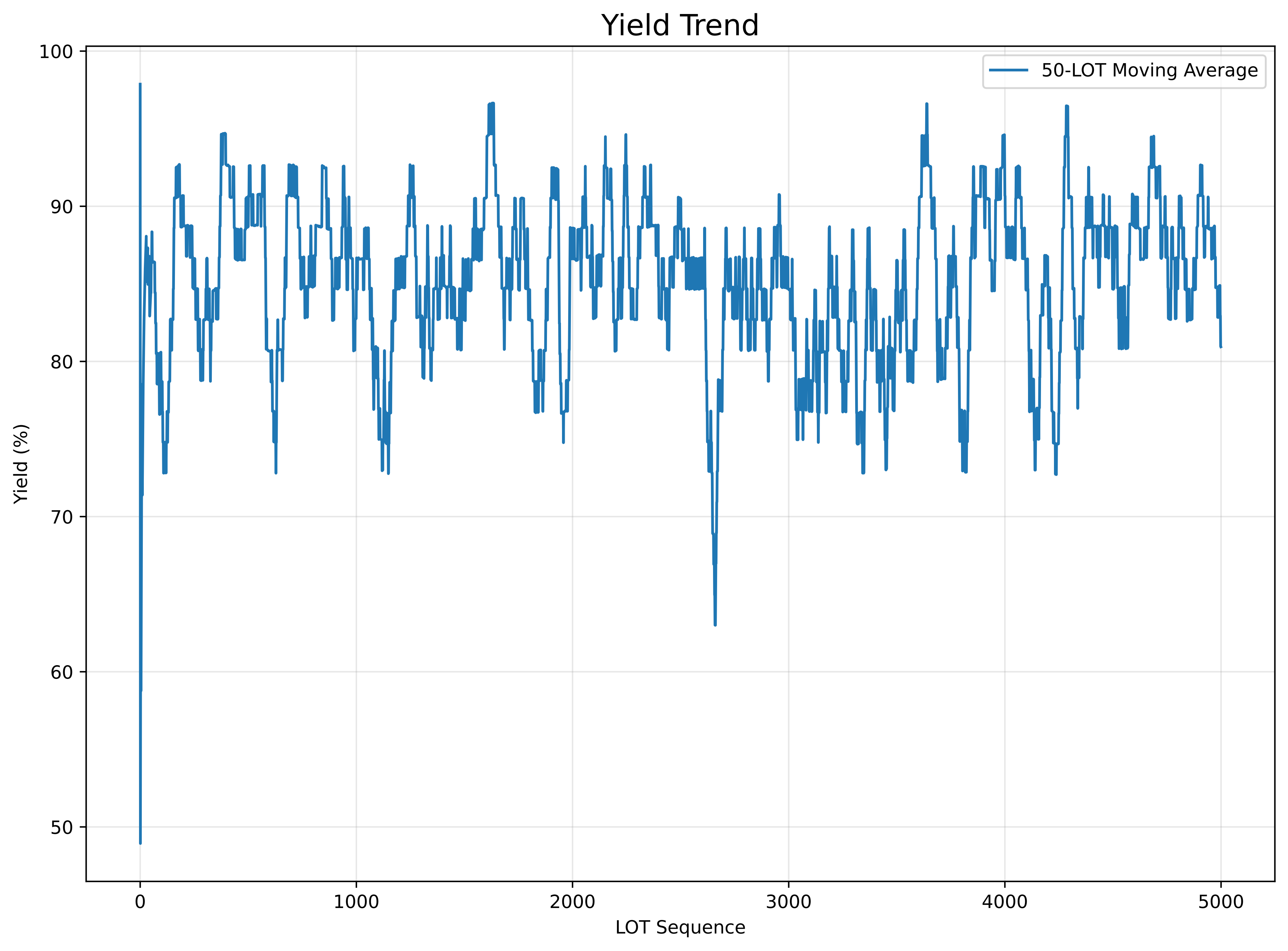
Defect Breakdown

- Delamination: 679 LOTs (13.58%)

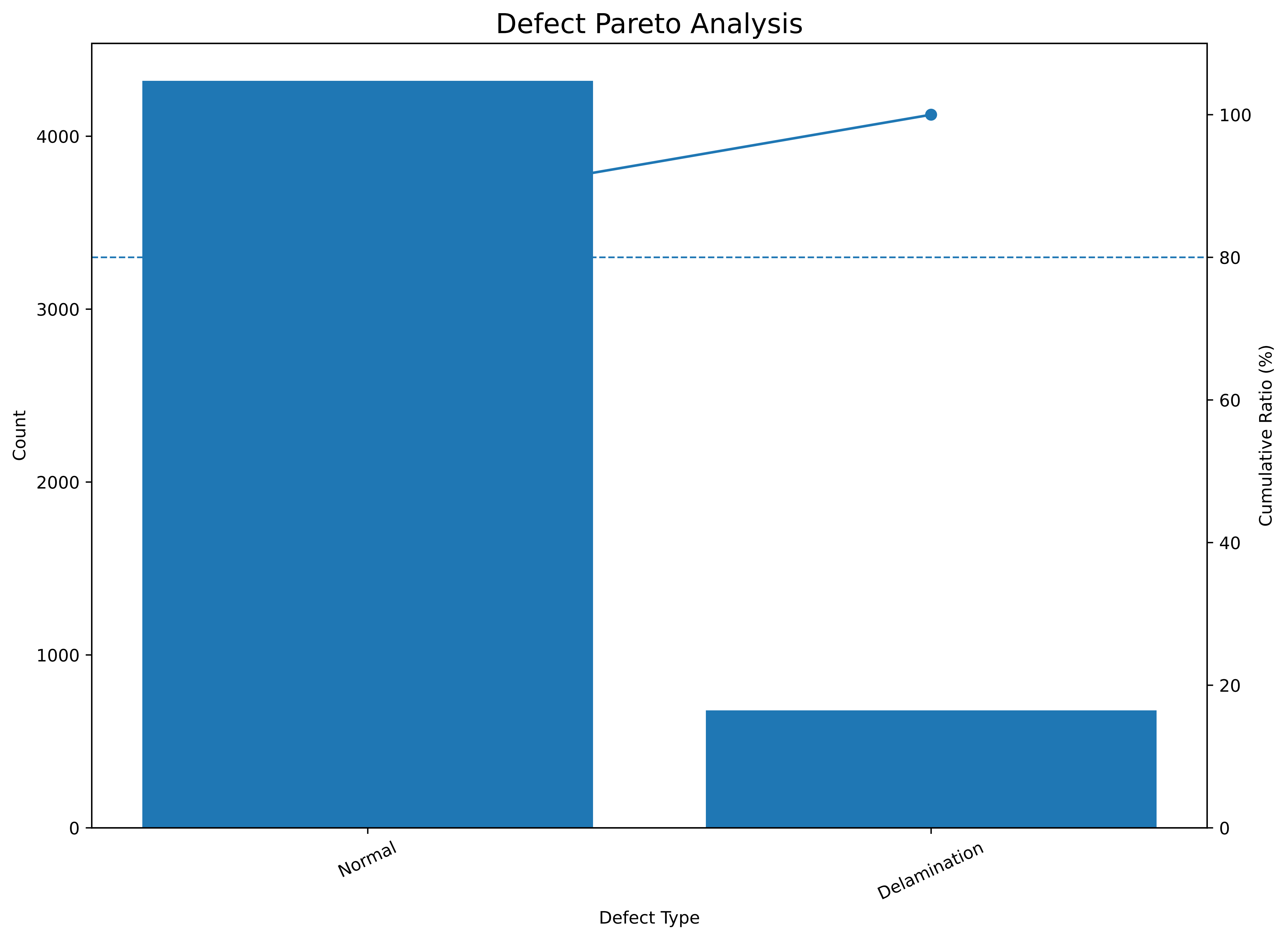
Conclusion

Based on the current 5,000-LOT simulation, the expected overall yield is 85.11% and the expected defect rate is 13.58%. The largest defect category is Delamination, 679 LOTs.

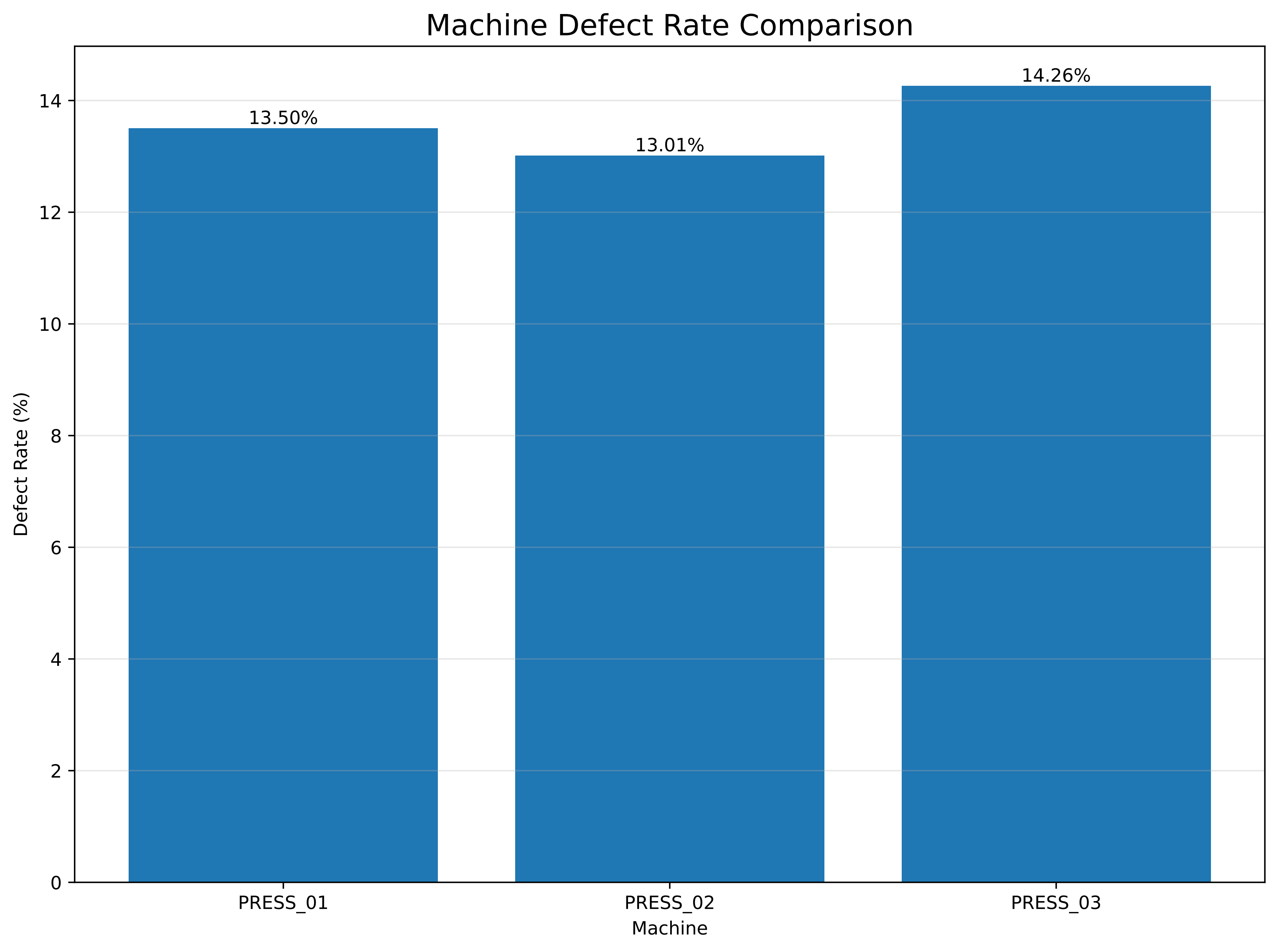
Note: The values above are simulation results generated from the current process model, not actual production forecasts.



Yield Trend
50-LOT Moving Average
Yield (%)
LOT Sequence
100
90
80
70
60
50
0
1000
2000
3000
4000
5000



Defect Pareto Analysis
Count
Cumulative Ratio (%)
Defect Type
Normal
Delamination
0
1000
2000
3000
4000
0
20
40
60
80
100


Machine Defect Rate Comparison
13.50%
13.01%
14.26%
Defect Rate (%)
0
2
4
6
8
10
12
14
PRESS_01
PRESS_02
PRESS_03
Machine

Process Capability Summary

Parameter	Mean	Std	LSL	USL	Index	Value	Assessment
CZ_Roughness	349.941	6.775	320.0	380.0	Cpk	1.473	Capable
Total_Thickness	32.001	0.368	30.0	34.0	Cpk	1.811	Excellent
Peel_Strength	499.781	20.004	350.0	nan	Cpl	2.496	Excellent
ABF_Roughness	349.817	6.701	300.0	400.0	Cpk	2.478	Excellent

Assessment guide: Excellent >= 1.67, Capable >= 1.33, Marginal >= 1.00, Improvement Required < 1.00

Cpk is used for two-sided specifications. Cpl or Cpu is used for one-sided specifications.

Quality Parameter Summary

Parameter	count	mean	std	min	25%	50%	75%	max
CZ_Roughness	5000.0	349.941	6.775	321.32	345.29	349.885	354.512	375.94
Total_Thickness	5000.0	32.001	0.368	30.57	31.752	32.004	32.248	33.281
Peel_Strength	5000.0	499.781	20.004	421.68	486.328	499.73	513.32	572.46
ABF_Roughness	5000.0	349.817	6.701	326.59	345.29	349.66	354.26	375.01
Yield	5000.0	85.105	33.746	0.0	97.72	98.34	98.85	100.0